

A Glossary of GRADE terms

The descriptions of the terms provided here are specific to how we use the terms in the context of Core GRADE

Instructions for Using the Glossary

- The glossary is divided into thematic categories.
- The content page below shows the categories and terms which are hyperlinked.
- Additionally, each category features a ***list of terms at the top of the page***, all of which are hyperlinked.
 - To navigate to a specific term's definition, hold down the ***Ctrl key (or Command key on Mac)*** and click on the term.
- Some terms are relevant to multiple categories and appear more than once.
 - In such cases, the term is defined only once in its primary section.
 - Subsequent mentions include ***a reference number*** that directs you to the section where the term is first defined.
 - To navigate to the ***primary definition***, hold down ***Ctrl (or Command)*** and click on the reference number.
- Within the definitions:
 - Words that are *italicized* and ***bold*** are meant for emphasis.
 - Terms that are ***bold and blue*** are hyperlinked and will take you to the original section where the term is defined.
- ***Use these hyperlinks and references for efficient navigation and comprehension of the glossary.***

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1. General concepts in Core GRADE

Core GRADE approach

Certainty of evidence; GRADE guidance, Plain Language Summary, Strength of Recommendations, Structured Clinical Question (PICO), Summary of Findings table

Core GRADE approach: *The essential principles and practices of GRADE in rating the certainty of evidence and moving from evidence to recommendations in comparisons of a single intervention versus a single comparator. Core GRADE does not include non-essential principles and practices with little additional benefit in generating trustworthy ratings of certainty and trustworthy recommendations.*

1.1 Certainty of evidence (synonym, “quality of evidence”): The extent to which the evidence allows trustworthy inferences regarding the impact of an intervention on a particular patient-important outcome. GRADE classifies certainty into four categories: high, moderate, low, and very low. Core GRADE asks users to specify a threshold (the null or the minimal important difference) and to rate their certainty that the true effect on an outcome lies on one side or other of the null or on one side or other of the minimal important difference (i.e. represents an important or unimportant effect).

1.2 GRADE guidance: A document providing suggestions for optimal use of GRADE in rating certainty of evidence and moving from evidence to recommendations in systematic reviews and clinical and systems guidance documents produced by a group of experts involved in the development, improvement and implementation of the GRADE approach and with extensive experience in teaching the GRADE approach.

1.3 Plain Language Summary: A statement, typically in a few words, that captures the essence of the findings of a comparison of intervention and comparator. **Summary of Findings (SoFs)** tables include plain language summaries in the last column. GRADE guidance includes wording suggestions for plain language summaries.

1.4 Strength of Recommendations: GRADE specifies two categories of strength of recommendations, **strong or conditional (synonym, “weak”)**. Core GRADE provides strong recommendations when all or almost all fully informed individuals would make the same choice for or against an intervention and a conditional recommendation when the majority of individuals would choose the suggested option, but an appreciable minority would – typically because of different values and preferences – choose the alternative.

1.5 Structured Clinical Question (PICO): The structured clinical question has four components: Patient, Intervention, Comparator, and Outcomes, frequently abbreviated as PICO.

1.6 Summary of Findings Table: A table presenting the key results of a systematic review typically including the study design of the included studies, the number of studies and patients, and for each patient-important outcome the relative effects, absolute effects, certainty of evidence, and a plain language summary.

2. Key terms related to guidelines and recommendations

Clinical Practice Guideline, Critical outcomes, Direction of Recommendation, Health Technology Assessment, Important outcome, Individual Patient Perspective, Strength of Recommendations, Systems Perspective synonyms: (Public Health Perspective, Population Perspective, Societal Perspective)

2.1 Clinical Practice Guideline: Structured recommendations regarding optimal clinical practice, ideally produced by a multidisciplinary panel including at minimum content area experts, methodologists, patient partners, and clinicians whose focus is providing patient care.

2.2 Critical outcomes: Outcomes that are very likely to have a major bearing on the strength or direction of a recommendation.

2.3 Direction of Recommendation: Recommendations can be either in favor of the intervention or in favor of the comparator.

2.4 Health Technology Assessment (HTA): HTA is a multidisciplinary process that uses explicit methods to determine the value of a health technology at different points in its lifecycle. The purpose is to inform decision-making in order to promote an equitable, efficient, and high-quality health system. (*O'Rourke, Oortwijn, Schuller, & the International Joint Task Group, 2020, p. 187*)

2.5 Important outcomes: Outcomes that patients value but may or may not have a major impact on the direction or strength of recommendations.

2.6 Individual patient perspective: A guideline panel or health technology assessment report takes a patient perspective when it focuses on all outcomes relevant to that patient and no outcomes not relevant to the patient. Outcomes relevant to patients include longevity, function, pain, quality of life, costs incurred by the patient, and burdens and adverse effects (synonym, "harms") of interventions. This perspective would not typically include costs born by the state, the acceptability of the intervention in the wider community, or equity issues.

2.7 Strength of Recommendations: *Refer to 1.4*

2.8 Systems perspective (synonyms "public health perspective" and "population perspective" "societal"): A guideline panel or health technology assessment report takes a systems perspective when it focuses on outcomes relevant to systems or public health decision makers. These would include issues of opportunity costs as a systems level, cost effectiveness, the acceptability and feasibility of the intervention in the community, and issues of equity.

3. Key concepts in Study Designs and Evidence Synthesis Methods

Case Series, Health Technology Assessment, Intervention and Comparator, Meta-analysis, Network Meta-analysis, Observational Studies (Non-Randomized Studies of Interventions – NRSI), Paired comparisons, Point Estimate, Pooled estimate, Randomized Controlled Trial, Structured Clinical Question (PICO); Systematic Review.

3.1 Case Series (Single Arm Study): A study in which all patients receive the same management strategy is typically referred to as a case series when the number of patients is modest. When the number is large, it is sometimes called a cohort study, but this can be confusing because the term 'cohort' is more commonly used to describe a non-randomized study in which some patients receive an intervention and others receive a comparator. The term "single-arm study" is more fitting for all such situations.

3.2 Health Technology Assessment (HTA): [Refer to 2.4](#)

3.3 Intervention and Comparator: Intervention and comparator represent alternative strategies for managing a health problem or concern. When one option is standard care or non-use of a particular intervention, we designate that alternative as the comparator and the other the intervention. When we are comparing two specific management strategies the designation of intervention or comparator is often arbitrary.

3.4 Meta-analysis: An analysis that provides a single pooled estimate of effect across studies as the difference ratio of the proportion of patients with an outcome (**relative effects** for **binary or yes-no outcomes**), or the difference in proportion of patients with an outcome (**absolute effect** or **risk difference** for **binary outcomes**) or the **mean difference** (for **continuous outcomes**).

3.5 Network Meta-analysis: A statistical technique that allows simultaneous comparison of two or more interventions and generates and estimate of effect of every possible pair of the interventions under consideration.

3.6 Observational Studies (Non-Randomized Studies of Interventions – NRSI): Studies in which patients receive the intervention or comparator according to their preference, their health care providers preference, some combination, or happenstance rather than randomization or quasi-randomization.

3.7 Paired comparisons: A comparison of a single intervention (e.g. corticosteroids plus standard of care) versus a single alternative (synonym, "comparator") on a patient-important outcome of interest.

3.8 Point Estimate: The single best estimate of a true effect. In single studies, the relative effect or absolute difference between values of an outcome in intervention and control groups. In a meta-analysis the pooled estimate of such differences.

3.9 Pooled Estimate: The single best estimate of effect from a meta-analysis coming from statistically combining the results of the individual studies.

3.10 Randomized Controlled Trial (RCT): A study in which patients are allocated to intervention or comparator according to a random process, typically a computer-generated random number table.

3.11 Structured Clinical Question (PICO): [Refer to 1.5](#)

3.12 Systematic Review: A summary of the best available evidence addressing a clinical question of patient management using a structured approach that includes explicit statement of the study question, a comprehensive search, and uses a structured approach to optimally synthesise evidence, assess evidence certainty, and present the key findings.

4. Statistical Terms and Measures

Absolute effects, Baseline Risk, Binary Outcomes, Confidence Interval, Continuous Outcomes, Dose-response gradient, Effect modification, Event Rate, Hazard Rate, Hazard Ratio, Interaction, Mean Difference, Meta-Analysis, Meta-regression, Odds ratio, Paired comparisons, Point Estimate, Pooled estimate, Relative Effects, Relative risk reduction(RRR), Relative odds reduction, Relative hazard reduction, Risk Difference (RD), Risk ratio (RR), Standardized mean difference (SMD), Subgroup Effect, Test of interaction, Time to Event (Survival), Weighted Mean Difference

4.1 Absolute effects: For binary outcomes, the difference in the proportion of patients experiencing an event in the treatment group versus the proportion of patients experience that event in the control group. We refer to this difference as the **risk difference**. For continuous outcomes the mean difference between the result in the intervention group and the result in the control group.

4.2 Baseline Risk: For binary outcomes, the risk or incidence of adverse outcomes in the comparator group (**comparator group risk**).

4.3 Binary Outcomes: Outcomes that can take one of two values (yes or no) that capture the presence or absence of an event such as death, stroke, or myocardial infarction.

4.4 Confidence Interval: In frequentist statistical analysis the plausible range of values in which the true effect of intervention versus comparator is likely to lie. The equivalent term in Bayesian analysis is the **credible interval**.

4.5 Continuous Outcomes: Outcomes that can take any of a range of values, typically from very low or zero to very high. These include age, many laboratory values, durations of illness or hospitalizations. We often divide outcomes that by their nature are continuous into categories (for age, years; for duration of illness or time in hospital, days, for certainty of evidence high, moderate, low or very low) or even binary (old or young).

4.6 Dose-response gradient: The existence of a relationship such that the greater the dose or amount of an intervention the greater the response on a particular outcome.

4.7 Effect Modification: An effect modification exists when a patient characteristic (such as age or sex), or an intervention characteristic (such as dose or mode of delivery) results in a different effect of an intervention. Synonyms include **subgroup effect** and **interaction**.

4.8 Event rate (risk, incidence): The frequency with which an event occurs in a population over a particular period of time without considering when the event happens during that period.

4.9 Hazard Rate: The statistical definition of hazard is hard to get one's mind around. Technically, it's been defined as the limit of the number of events per unit time divided

by the number at risk as the time interval approaches 0. One way this might be expressed is the

instantaneous likelihood of an event. Either way, it sounds obscure. Thinking about it as the likelihood of an event, such as death or stroke, occurring over a specified period of time conveys the basic idea. The hazard rate can vary over time. For example, after a major surgery, deaths may happen frequently in the first days after surgery, considerably less often in the first month, and then much less frequently after the first month.

4.10 Hazard Ratio: A hazard ratio is the ratio of one hazard rate to another, typically the hazard in an intervention group compared to the hazard rate in the control group. It is typically used in time to event (survival) analyses that take into account not only if an event happens but when it happens. The hazard ratio assumes that the relative likelihood of events happening in one group, or another stays more or less the same over time. If that is not true, a single hazard ratio does a bad job of capturing the possible impact of treatment over time. For instance, if patients are more likely to have a stroke or die after cerebrovascular surgery than patients not receiving surgery in the first month after surgery but subsequent less likely to die or have a stroke the hazard ratio will be greater than 1 in the first month and less than 1 thereafter. If the hazard ratio is similar over time (we say it meets the *proportional hazard assumption*) then you can think of it as the relative risk averaged over time.

4.11 Interaction: An interaction between a patient characteristic (such as age or sex), or an intervention characteristic (such as dose or mode of delivery) and a treatment effect exists when that effect differs in different patient groups or interventions. Synonyms include **effect modification** and **subgroup effect**.

4.12 Mean Difference: For continuous variables, ideally with a reasonably normal distribution, difference between the mean value in the intervention group and the mean value in the control group.

4.13 Meta-analysis: [Refer to 3.4](#)

4.14 Meta-regression: In meta-analysis, an analysis in which the independent variable(s) are one or more possible determinants of the relative effect, and the dependent variable is the relative effect associated with an intervention. Independent variables are typically patient characteristics (such as age or sex) but can also be intervention characteristics (such as dose) or risk of bias.

4.15 Odds ratio: The relative odds of an effect in the intervention group to an effect in the comparator group. (**Synonym relative odds**)

4.16 p-values (hypothesis testing): The hypothesis testing approach to statistical inference operates within what is called a frequentist framework, posits a null hypothesis, and uses the data to challenge that hypothesis. If the data is sufficiently inconsistent with the hypothesis, and thus the probability that results as or more extreme would be observed if the null hypothesis were correct (the p-value) is sufficiently low, one rejects the hypothesis. Use of p-values for making inferences about treatment effects has been largely rejected because of a number of problems. These include peoples' tendency to interpret a p-value greater than the typical threshold of 0.05 as indicating that the intervention has no effect. The correct interpretation is that the intervention may still have an important effect, but the data

has

failed to provide convincing evidence that it does. The alternative to hypothesis testing that we use in Core GRADE is called **estimation** and relies on **point estimates** and **confidence intervals**. The only time you will see p-values in Core GRADE is in the context of **interaction tests** used in **subgroup analysis**.

4.17 Paired comparisons: *Refer to 3.7*

4.18 Point Estimate: *Refer to 3.8*

4.19 Pooled Estimate: *Refer to 3.9*

4.20 Relative Effects: The ratio of an effect in the intervention group to an effect in the comparator group. Relative effects can be **relative risks** (ratios of risks or **risk ratios**), **relative odds** (more commonly referred to as **odds ratios**), or **hazard ratios** (ratios of two **hazard rates** in a time to event analysis). One can consider the magnitude of a treatment effect as a **relative risk reduction**, a **relative odds reduction**, or a **relative hazard reduction**.

4.21 Relative risk reduction (RRR): *Refer to 4.20*

4.22 Relative odds reduction: *Refer to 4.20*

4.23 Relative Hazard reduction: *Refer to 4.20*

4.24 Risk Difference (RD): *Refer to 4.1*

4.25 Risk ratio (RR): The relative risk of an effect in the intervention group to an effect in the comparator group. (**Synonym relative risk**)

4.26 Standardized mean difference (SMD) (Effect size): Typically used in meta-analysis of patient-reported outcomes when studies use different instruments to measure the same construct. The SMD represents the mean effect in each study divided by its standard deviation. The pooled estimate (the SMD) is then a value in standard deviation units.

4.27 Subgroup Effect (synonyms effect modification and interaction): These three terms refer to estimates of relative effect that may differ according to patient characteristics (old versus young in which case age is the effect modifier, male versus female in which case gender is the effect modifier) or intervention characteristics (high versus low dose, shorter versus longer duration). A subgroup effect, effect modification, or an interaction exists when relative effects do actually differ across patient or intervention characteristics.

4.28 Test of interaction: A statistical test with the null hypothesis that there is an identical effect of treatment across patient characteristics (such as age or sex) or intervention characteristics (such as dose or mode of delivery). Low p-values of the test of interaction suggest that chance is an unlikely explanation for differences in effect observed across patient or intervention characteristics.

4.29 Time to Event (Survival) Analysis: A statistical analysis typically comparing an intervention and control that considers not only when an event happens but how frequently it happens over time.

4.30 Weighted Mean Difference (WMD): In a meta-analysis in which each study has measures a construct using the same measurement instrument or units and represents the pooled estimate as a mean difference between intervention and comparator.

5. GRADE Domains for rating down and rating up and related terminology

5a. Imprecision

Absolute effects, Confidence Interval, Magnitude of Effect, Minimal Important Difference (MID), Non-null Effect, Optimal Information Size (OIS), Point estimate, Pooled estimate, Relative Risk Reduction (RRR), Risk Difference (RD), Target of Certainty Rating, Threshold, Weighted Mean Difference (WMD).

5a. Imprecision: The extent of random error associated with an estimate of effect as reflected in the confidence interval around the estimate. The wider the confidence interval the greater the imprecision.

5.1 Absolute effects: *Refer to 4.1*

5.2 Confidence Interval: *Refer to 4.4*

5.3 Magnitude of Effect: An underlying continuum from no difference between intervention and comparator to a very large difference between intervention and comparator represented by relative or absolute effects or mean differences. We often develop categories for this magnitude which in Core grade are ***non-zero, little or no difference, or an important difference***. Important difference can have additional categories for *small, medium, and large*.

5.4 Minimal Important Difference (MID): The smallest effect of an intervention on an outcome of interest that patients would consider important.

5.5 Non-null Effect (Non-zero effect): Any effect other than the ***null***; that is, any effect other than a relative effect of 1.0 or an absolute effect of 0.

5.6 Optimal Information Size (OIS): When we see implausibly large effects emerge from a meta-analysis, we are sceptical: such effects often prove, with further evidence, to be overestimates. The smaller the sample size and number of events, the more likely the overestimate and the more likely it is to be a large overestimate. To guard against overly sanguine ratings of certainty, when confidence intervals do not otherwise mandate rating down for imprecision, we invoke the OIS. The OIS is the sample size that we would require when doing a power calculation for a single trial choosing a delta that represents a modest effect (typically a 20% or 25% reduction in relative effect). If the total sample size in the meta-analysis is greater than the OIS we do not rate down for imprecision; if less, we do rate down.

5.7 Point Estimate: *Refer to 3.8*

5.8 Pooled Estimate: *Refer to 3.9*

5.9 Relative Risk Reduction (RRR): *Refer to 4.20*

5.10 Risk difference: *Refer to 4.1*

5.11 **Target of Certainty Rating:** An alternative term for *that in which we are rating our certainty*. In Core GRADE alternative targets include ***little or no effect***, ***non-zero effect***, and ***important effect***.

5.12 **Threshold:** Any value that divides a continuum into categories. In GRADE, common thresholds include **non-serious**, **serious** and **very serious concerns** in factors that may lead to rating down certainty of evidence; **strong** versus **conditional recommendations**; the **null effect** and the **minimally important difference (MID)**.

5.13 Weighted Mean Difference (WMD): *Refer to 4.30*

5. GRADE Domains for rating down and rating up and related terminology

5b. Inconsistency

A priori hypothesis, Credibility of subgroup analysis, Effect modification, I^2 statistic, Interaction, Subgroup Effect, Test of interaction

5b. Inconsistency: The extent to which results differ from study to study. The greater the variability in results, the greater the inconsistency. Sometimes referred to as *variability in results*, and particular when referring to statistical variability, *heterogeneity*.

5.14 A priori hypothesis: A hypothesis, typically about the direction of effect, that investigators generate before beginning a study.

5.15 Credibility of sub-group analysis: Inferences are credible to the extent that we believe they are likely to be accurate and represent some underlying truth. **Certainty or quality of evidence** represent the credibility of our inferences regarding the evidence. In the **ICEMAN** instrument addressing subgroup analysis, to the extent that we believe subgroup hypotheses or claims accurately identify true underlying effect modification those hypotheses or claims are credible; to the extent they do not, they are not credible.

5.16 Effect modification: [Refer to 4.7](#)

5.17 I^2 statistic: I^2 describes the percentage of the variability in effect estimates that is due to heterogeneity rather than sampling error (chance)

5.18 Interaction: [Refer to 4.11](#)

5.19 Subgroup effect: [Refer to 4.27](#)

5.20 Subgroup hypothesis: A hypothesis investigators generate to explain possible differences in effect within studies or between studies. These most typically involve patients (young versus old), interventions (high dose or low dose) and risk of bias (high versus low).

5.21 Test of Interaction: [Refer to 4.28](#)

5. GRADE Domains for rating down and rating up and related terminology

5c. Indirectness

Direct Evidence, Indirect Evidence, Non-Adherence (Non-Compliance), Patient-Important Outcome, Sub-optimal comparators, Surrogate Outcome, Target PICO

5c. Indirectness: The extent to which the patients, interventions, comparators and outcomes in the available studies differ from these PICO elements in the target PICO. Alternative terms include lack of **generalizability or inapplicability**.

5.22 Direct Evidence: Evidence that comes from studies that meet the **target PICO**. That is, the studies enrol the target population, compare the target intervention to the target comparator, and report on at least one of the target outcomes.

5.23 Indirect Evidence: There exist two types of indirect evidence. The first directly compares an intervention to a comparator of interest related to the target PICO but differing importantly in the patients, interventions, comparators or outcomes. The second involves indirect comparisons: we are interested in A versus B and infer the relative effects of A versus B from, for instance, comparisons of A versus C and B versus C.

5.24 Non-adherence (non-compliance): Patients failure to use an intervention as directed or recommended by a health care provider.

5.25 Patient-important outcome: An outcome that would, if influenced by the intervention, despite intervention-associated harms and burdens, lead the patient to consider the intervention. For instance, were an intervention to reduce hip fractures patients might consider the intervention even if that intervention were associated with appreciable harm or burden. In contrast, **were an intervention to improve bone density without any impact of longevity, fractures, or bone pain**, in the face of intervention-induced harms or burdens patients would not be interested in that intervention.

5.26 Sub-optimal comparator: A comparator that may mislead clinicians. The most common and problematic comparators are those chosen in an industry study to make a new treatment look better than it would against an optimal comparator.

5.27 Surrogate outcome: An outcome that, were it the only outcome to change, would not interest a patient in using an intervention with associated harms and burdens. For instance, were an intervention to reduce hip fractures patients might consider the intervention even if that intervention were associated with appreciable harm or burden (thus, a **patient-important outcome**). In contrast, were an intervention to improve bone density **without any impact of longevity, fractures, or bone pain**, in the face of intervention-induced harms or burdens patients would not be interested in that intervention (thus, a **surrogate outcome**).

5.28 Target PICO: The combination of the patients, interventions, comparators and outcomes that represent the question of interest in a systematic review, practice guideline, or health technology assessment.

5. GRADE Domains for rating down and rating up and related terminology

5d. Risk of Bias

Blinding, Concealment and Randomization, Confounder, Direction of Effect, Dose-response gradient, Intention-To-Treat analysis (ITT), Prognostic factors, ROBUST-RCT, Small study effects

5d. Risk of Bias: Failure to implement safeguards against bias (such as concealment, blinding, and loss to follow-up in randomized trials and adjusted analysis in NRSI) puts a study at risk of bias.

5.29 Blinding: A study is blinded when one or more of the following are unaware of individual patient allocation to intervention or comparator: patients, health care providers, data collectors, adjudicators of outcomes, statistical analysts, or those interpreting the results.

5.30 Concealment of Randomization: Randomization is concealed when the individual enrolling patients in a randomized trial is unaware of whether the next patient to be enrolled will be allocated to the intervention or the comparator.

5.31 Confounder: A prognostic factor for an outcome of interest that is not equally occurring in an intervention and control group. For instance, age, a prognostic factor for death, will become a confounder if the distribution of age differs in intervention and control groups and the outcome is mortality. One can think of randomization as a strategy for preventing a prognostic factor from becoming a confounder.

5.32 Direction of effect: Direction of effect refers to whether a point estimate favors an intervention or a comparator.

5.33 Dose-response gradient: The existence of a relationship such that the greater the dose or amount of an intervention the greater the response on a particular outcome.

5.34 Intention-To-Treat analysis (ITT): An analysis in which patients are analyzed in the groups to which they are randomized. However, authors use the term differently. Possibilities are i) All patients have been followed as planned for the outcome of interest, ii) All patients have not been followed but investigators impute results for those not followed, iii) ITT has to do with patients who are followed as planned being analyzed in the groups to which they are randomized and how one deals with loss to follow-up is a separate issue. Core GRADE uses the third approach.

5.35 Prognostic Factor (Risk Factor): A patient characteristic such as age, sex, or disease severity associated with the likelihood of an outcome such as stroke, myocardial infarction, or death. The term **risk factor** is more commonly used for prognosis of individual who are otherwise well (for instance the risk of a myocardial infarction in individuals who appear well but may be at increased risk because of age, lipid status, or family history) and **prognostic factor** for the likelihood of bad events in someone who already has a disease or condition (such as recurrent myocardial infarction). However, in both situations one can refer to factors influencing the likelihood of subsequent event as prognostic factors.

5.36 ROBUST-RCT: An instrument to measure risk of bias in individual RCTs the development of which was completed in 2024. It is the preferred instrument in Core GRADE.

5.37 Small study effects: Small study effects occur when small studies in a systematic review show different results than larger studies, typically larger effects than the large studies. Explanations can be publication bias; small studies results are more influenced by bias that may sometimes be difficult to detect; and small studies follow treatment protocols more rigorously.

5. GRADE Domains for rating down and rating up and related terminology

5e. Publication Bias

Funnel Plot, Selective Reporting

5e. Publication Bias (Dissemination bias): Publication bias occurs when the results of a study influence its likelihood of being published. Typically, studies are less likely to be published if their results fail to show a difference between intervention and control groups. Publication bias for a particular outcome will bias the results of a systematic review and meta-analysis of that outcome.

5.38 Funnel Plot: An approach to explore publication bias or small study effects. On the x-axis the plot has the magnitude of effect and on the y-axis the sample size or random error represented by the statistical variance or imprecision. If the results of the small studies are not symmetrically distributed around the estimates of the large studies, it suggests small study effects and the possibility of publication bias.

5.39 Selective Reporting: Selective reporting occurs when, within a study, the results for a particular outcome influence its likelihood of being reported. When an outcome is not reported at all, selective reporting is a form of **publication bias** (the information for that outcome unavailable and will therefore likely bias the results of a systematic review and meta-analysis). Reporting results but less completely (for instance for some outcomes giving point estimates and confidence intervals, but for others stating only “non-significant”) is a less extreme form of selective reporting.

5. GRADE Domains for rating down and rating up and related terminology

5f. Related terms

Continuum of certainty, High certainty, Low certainty, Moderate certainty, Non-Serious concerns, Rating down certainty of evidence, Rating up certainty of evidence, Serious concerns, Very low certainty, Very Serious concerns

5.40 Continuum of certainty: Certainty of evidence constitutes a continuum from extremely low certainty to absolute certainty. GRADE breaks this continuum into categories of high, moderate, low and very low certainty. It is worth, however, bearing in mind that the underlying certainty may be near the threshold between any two adjacent categories.

5.41 High certainty: We are very confident that the true effect lies on one side of a threshold or in a particular range.

5.42 Low certainty: Our confidence that the true effect lies on one side of a threshold or in a particular range is limited.

5.43 Moderate certainty: We are moderately confident that the true effect lies on one side of a threshold or in a particular range.

5.44 Non-Serious concerns: Limitations in imprecision, inconsistency, risk of bias, indirectness, or publication bias, not important enough to warrant, in and of themselves, rating down. Such concerns may in the final gestalt evaluation of uncertainty, tip an uncertain overall rating over from a higher to lower category.

5.45 Rating down certainty of evidence: Our **certainty** or judgement of **quality** of evidence decreases when a body of evidence is characterized by serious limitations as a result of imprecision, inconsistency, risk of bias, indirectness, or publication bias.

5.46 Rating up certainty of evidence: Our **certainty** or judgement of **quality** of evidence in a body of evidence from **non-randomized studies of interventions (NRSI, observational studies)** may increase when the intervention effect is **large or very large** or when there is a **dose-response gradient**.

5.47 Serious concerns: Concerns serious enough to warrant rating down the certainty by one level due to limitations in imprecision, inconsistency, risk of bias, indirectness, or publication bias.

5.48 Very low certainty: We are very uncertain whether the true effect lies on one side of a threshold or in a particular range.

5.49 Very Serious concerns: Concerns serious enough to warrant rating down the certainty by two levels due to limitations in imprecision, inconsistency, risk of bias, indirectness, or publication bias.

6. Evidence-to-Decision Factors

Acceptability, Benefit and Harm, Burdens, Costs (Resource Use), Equity, Feasibility, Minimal Important Difference, Values and Preferences

6.1 Acceptability: The extent to which patients, caregivers, health care providers and the community find implementation of an intervention bearable, tolerable, or supportable.

6.2 Benefit: The potential desirable outcomes that may result from an intervention.

Harm: The potential undesirable outcomes that may result from an intervention.

6.3 Burdens: Burdens have to do with what has been called the “work” of being a patient and GRADE guidelines have referred to as practical issues. This work includes, medication frequency and route, tests and visits, procedures and devices, coordination of care, recovery and adaptation, managing dependencies, directions regarding diet, exercise, and health habits, impact on work and social life, and any physical or emotional distress that may come with managing all these issues. (*Heen, Vandvik, Brandt, et al., 2021*)

6.4 Costs (Alternative term resource use): The resources required to implement an intervention expressed in monetary terms.

6.5 Equity: The impact of an intervention on how the desirable and undesirable consequences of an intervention affect different constituencies eligible for the intervention.

6.6 Feasibility: The extent to which a patient, caregiver, or health care system are able to implement an intervention.

6.7 Minimal Important Difference (MID): *Refer to 5.4*

6.8 Values and Preferences: Values and preferences broadly refers to the set of issues (e.g. convenience, restrictions, risks, uncertainty of benefit, burden on others, distrust of medicine etc.) that make a course of action more or less desirable for the participants (typically, the patient). These issues may originate from many sources (e.g., participants’ **perspectives, beliefs, expectations, priorities, and goals** for their health and life, **situational factors**, an individual’s **predispositions and aversions, cultural norms and narratives**, the **wishes of significant others**, etc.) and be expressed in many ways, but they characteristically offer stronger or weaker reasons for or against an action and these reasons are matters for deliberation in shared decision making.

Reference:

O'Rourke B, Oortwijn W, Schuller T, the International Joint Task Group (2020). The new definition of health technology assessment: A milestone in international collaboration. *International Journal of Technology Assessment in Health Care* 36, 187–190. <https://doi.org/10.1017/S0266462320000215>

Heen, A. F., Vandvik, P. O., Brandt, L., et al. (2021). A framework for practical issues was developed to inform shared decision-making tools and clinical guidelines. *Journal of Clinical Epidemiology*, 129, 104–113. <https://doi.org/10.1016/j.jclinepi.2020.10.002>